Chemistry SW2 – Structure of Atoms

1) Indicate the number of protons, neutrons, and electrons for the following isotopes:

Isotope	Element Name	Number of Protons	Number of Neutrons	Number of Electrons
^{2}H	Hydrogen (Deuterium)			
^{3}H	Hydrogen (Tritium)			
^{12}C	Carbon			
^{27}Al	Aluminum			
^{65}Cu	Copper			
^{204}Pb	Lead			
^{235}U	Uranium			

2) What is a chemical element? What characteristic defines an element?

3) What information can be obtained from the atomic number?

4) What do isotopes of a chemical element have in common, and how do they differ?

5) Name a reason why the relative atomic masses in the periodic table are not whole numbers.

6) A (fictional) element E has an average nucleon number of 88.20. It is made up of the isotopes ^{85}E and ^{89}E. What is the ratio of the isotopes?

7) a) What is the isotope effect?
b) For which element is the isotope effect particularly pronounced, and why?

8) Which of the following statements applies to all atoms, including the hydrogen atom?

They always contain the following elementary particles in the nucleus:
a) one or more protons
b) one or more neutrons
c) one or more electrons
d) neutrons and protons in a 1:1 ratio
e) neutrons and protons

9) Which of the following statements is correct? The nuclides ^{3}H and ^{3}He…

a) …have the same number of neutrons
b) …have a different number of electrons
c) …are isotopes of the same chemical element
d) …are isobars

10) The radioactive ^{14}C decays to ^{14}N. What decay mechanism could be responsible? What kind of radiation (particles) is released?

11) Why is carbon-14 used for radiocarbon dating and not, for example, radioactive ^{11}C?

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1) Indicate the number of protons, neutrons, and electrons for the following isotopes:

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^{3}H	Hydrogen (Tritium)	1	2	1
^{12}C	Carbon	6	6	6
^{27}Al	Aluminum	13	14	13
^{65}Cu	Copper	29	36	29
^{204}Pb	Lead	82	122	82
^{235}U	Uranium	92	143	92

2) What is a chemical element? What characteristic defines an element?

A chemically indivisible substance. All atoms of a chemical element have the same number of protons in the nucleus (= the same atomic number).

3) What information can be obtained from the atomic number?

a) the number of protons
b) the number of neutrons
c) the number of electrons
d) the number of isotopes

4) What do isotopes of a chemical element have in common, and how do they differ?

They have different numbers of neutrons but the same electron shell. As a result, they behave almost identically in chemical reactions.

5) Name a reason why the relative atomic masses in the periodic table are not whole numbers.

Many natural elements are mixtures of isotopes. The atomic masses of naturally occurring elements therefore depend significantly on the percentage composition of their isotopes.

6) A (fictional) element E has an average nucleon number of 88.20. It is made up of the isotopes ^{85}E and ^{89}E. What is the ratio of the isotopes?

Proportion of ^{85}E: x; proportion of ^{89}E: 1 - x
x * 85 + (1 - x) * 89 = 88.2

$-4x = -0.8 \Rightarrow x = 0.2$
$\rightarrow$ 20% ^{85}E, 80% ^{89}E

7) a) What is the isotope effect?
b) For which element is the isotope effect particularly pronounced, and why?

a) Isotopes have different masses, so their reaction rates can differ slightly.
b) The effect is particularly significant for hydrogen because its isotopes have relatively large mass differences.

8) Which of the following statements applies to all atoms, including the hydrogen atom?

They always contain the following elementary particles in the nucleus:
a) one or more protons ✔
b) one or more neutrons ✗
c) one or more electrons ✗
d) neutrons and protons in a 1:1 ratio ✗
e) neutrons and protons ✗

9) Which of the following statements is correct? The nuclides ^{3}H and ^{3}He…

a) …have the same number of neutrons ✗
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c) …are isotopes of the same chemical element ✗
d) …are isobars ✔

10) The radioactive ^{14}C decays to ^{14}N. What decay mechanism could be responsible? What kind of radiation (particles) is released?

Radioactive carbon-14 decays to nitrogen-14 through beta decay (β-radiation). A neutron in the nucleus transforms into a proton, releasing an electron (and an antineutrino, not required knowledge here).

11) Why is carbon-14 used for radiocarbon dating and not, for example, radioactive ^{11}C?

The method relies on the decay of carbon-14, which has a half-life of 5730 years. The ratio to the stable isotope ^{12}C provides information about the age of a sample. The half-life of ^{11}C is around 20 minutes, making it unsuitable for dating purposes.